

Lab sessions

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1 Lab Practice: Problem Introduction

Step 1: Selecting an Insulating Antiferromagnet

- **Selection Criteria:**

- Choose a material with ≤ 20 **atoms** per cell to minimize computational cost.
- **Exclude Lanthanides** due to technical complexities in modeling f -orbitals.

- **Where to find Antiferromagnetic Materials:**

- *Physical Review B* **111**, 144406 — **108**, 144413
- **MAGNDATA**: <https://cryst.ehu.es/magndata/>

- **Obtaining Experimental Crystal Structures:**

- Relevant literature and databases.
- **COD**: <https://www.crystallography.net/cod/>
- **MAGNDATA** (Magnetic CIF files).

Step 2: Structural vs. Magnetic Symmetry

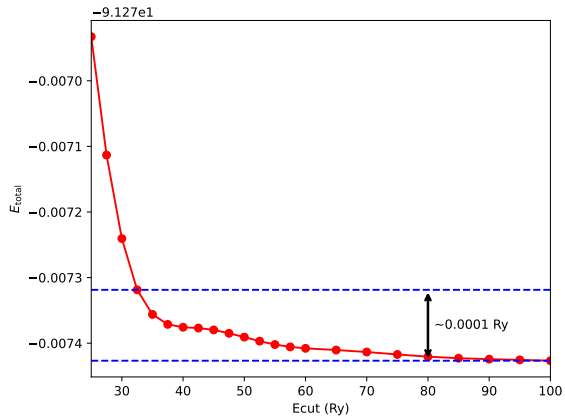
- **Symmetry Breaking:** Magnetic ordering often lowers the symmetry of the original crystal lattice.
- **The Magnetic Unit Cell:** To represent an antiferromagnetic state, you must typically define a **supercell** (two or more sublattices) to accommodate both spin-up and spin-down orientations.
- **Your Task:** Compare your experimental crystal structure with your magnetic model:
 - Does the magnetic order require a larger cell than the primitive chemical cell?
 - Identify the expansion (e.g., $1 \times 1 \times 2$ supercell) and explain the changes in symmetry.

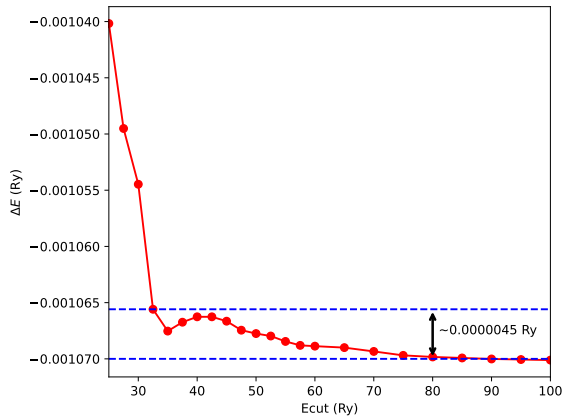
Tip: Use a visualization tool like VESTA to show the chemical vs. magnetic cells side-by-side.

Step 3: Convergence & Parameter Optimization

- **Parameters to Optimize:**
 - **Kinetic Energy Cutoff** (E_{cut})
 - **K-point Grid Density** (Brillouin zone sampling)
- **Compare Two Optimization Strategies:**
 - ① **Total Energy** (E_{tot}): Convergence of the absolute value.
 - ② **Energy Difference** (ΔE): Difference between two volumes (e.g., V_{exp} and $1.15 \times V_{exp}$).
- **Critical Analysis:** Discuss which strategy is more **computationally meaningful** and effective for your specific physical properties.

Total energy vs ecut



ΔE vs E_{cut} 

Step 4: Magnetic Phase Stability (Total Energy)

Task: Determine the Magnetic Ground State

- Compute total energies (E_{tot}) for three distinct states:
 - ① **Non-magnetic (NM)**: Spin-unpolarized calculation.
 - ② **Ferromagnetic (FM)**: Parallel spin alignment.
 - ③ **Antiferromagnetic (AFM)**: Anti-parallel alignment.
- **Complex AFM**: If the cell has > 2 magnetic sites, test different AFM configurations.
- **The $+U$ Effect**: Compare ΔE (energy differences) between standard **DFT** and **DFT+U** (Literature/Calculated U).

Scientific Inquiry

Why do magnetic stability and energy differences change so drastically with U ?

Step 5: projected density of state (PDOS) with Hubbard U (DFT+U)

- plot PDOS for spin up and down for antiferromagnetic (AFM) ordering.
- Perform a comparative study:
 - **Standard DFT** ($U = 0$)
 - **DFT+U** (Literature/Calculated U)

Key Analysis

- Compare the DFT and DFT+U PDOS
- Are spin-up and spin-down PDOS symmetric?
- How does U affect the localization of d electrons?

Step 6: Band Structure with Hubbard U (DFT+U)

Task Objectives

- Plot spin-resolved bands ($\uparrow$ and $\downarrow$) for **antiferromagnetic (AFM)** ordering.
- Perform a comparative study:
 - **Standard DFT** ($U = 0$)
 - **DFT+U** (Literature/Calculated U)

Key Analysis

- Compare the DFT and DFT+U Band Gap (with experiment)
- Are spin-up and spin-down bands symmetric?

Extra Task (Optional)

Calculation of Hubbard U from First Principles

If resources allow, use the **Density Functional Perturbation Theory (DFPT)** linear response method instead of empirical values.

- **Why?** Avoids the "fitting" bias and ensures the U value is self-consistent with your specific crystal structure and pseudopotentials.
- **Method in QE:**
 - 1 Perform a standard ground-state SCF calculation.
 - 2 Use the `hp.x` code (Hubbard Parameters) to calculate the response to a localized perturbation.
 - 3 Solve the linear response equations to extract the effective U .
- **Result:** A U parameter that is *ab initio* and unique to your system.

Requirement

Note: This is computationally expensive and requires the HP package included in the standard QE distribution.

Final Task: Reporting Your Results

Deliverable: A Structured Technical Report

The report should document your full computational workflow:

- **Methodology & Convergence**

- **Visualization:** Structural and magnetic ordering (AFM) using VESTA/XCrySDen.
- **Setup:** Choice of Exchange-Correlation (PBE/PBEsol) and pseudopotentials.
- **Tests:** Convergence studies for `ecutwfc` and `K_POINTS`.

- **Electronic & Magnetic Properties**

- **Spectra:** Band structure and PDOS plots (comparing DFT vs. DFT+U).
- **Energetics:** Total energy comparison for NM, FM, and AFM phases.

- **Discussion & Conclusion**

- Analyze how the Hubbard U parameter impacts electron localization and the band gap in your material.

Advanced Tools and Tips

Building `pw.x` Input Files

- **Materials Cloud Input Generator:**
<https://qeinputgenerator.materialscloud.io/>
- **Atomic Environment Simulation (AES):** Use built-in modules for structure manipulation and script generation.

Band Structure & K-Path Selection

- **SeeK-path:** Automatically generate standardized k-paths for any crystal structure.
<https://seekpath.materialscloud.io/>

Projected Density of States (PDOS)

- Refer to the `PP/examples` directory in the QE source folder for `projwfc.x` templates.

Advanced Tools and Tips

Choosing Pseudopotentials

- **QE Legacy Tables:**

http://pseudopotentials.quantum-espresso.org/legacy_tables

- **SSSP Efficiency:** Recommended for high-throughput.

<https://legacy.materialscloud.org/discover/sssp/table/efficiency>

- **GBRV (Vanderbilt Ultrasoft):** Highly optimized set.

<https://www.physics.rutgers.edu/gbrv/>

- **Pseudo Dojo:** Standardized Norm-Conserving potentials.

<https://www.pseudo-dojo.org/>